Forecasting Budgetary Items in Türkiye Using Deep Learning

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July 2025
Working Paper No: 25/09

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Abstract

This study aims at forecasting the future behavior of budget variables, using Artificial Neural Network (ANN) and Deep Neural Network (DNN) techniques for Türkiye. Particularly, we focus on budget expenditures, tax revenues and their main components. Annual data were used and divided into two sub-periods: a training set (2002-2019) and a test set (2020-2022). Each fiscal item is estimated using relevant explanatory variables selected based on economic theory. We achieved good forecasting performance for main budget items using ANN and DNN methodologies. We found that most of the Mean Absolute Error (MAE) values fell within the acceptable range, an indicator of good prediction performance. Second, we see that the MAE values for public expenditures are lower than taxes. Third, estimating total tax revenues (aggregate data) performs better compared to subcomponents of taxes (disaggregated data). The opposite is the case for public expenditures.

JEL code: C53, H20, H50, H68

Keywords: Machine Learning, Deep Learning, Artificial Neural Network (ANN), Deep Neural Network (DNN), Budget Forecast, Government Spending, Tax Revenue

(*) We would like to thank the participants who attended the seminar held at the CBRT on March 25, 2024, for their valuable suggestions. We also thank the anonymous referee for their valuable comments.
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Non-Technical Summary

Using deep learning in the field of economics has gained great importance recently. Neural network models such as Artificial Neural Network (ANN) and Deep Neural Network (DNN) models have been found to be useful to predict the future values of a variable. These models have been used for the forecasting of several macroeconomic indicators such as GDP growth, inflation, exchange rates and stock returns. It is also common for this methodology to be used in fiscal policy, mainly by estimating government expenditures. Particularly, measuring government size and estimating government expenditures using functional classification of spending are the main focuses of new methodology in the fiscal policy side. Since the composition of public expenditures is important for economic growth, the role of public spending in shaping economic growth is the focus of these studies.

The aim of fiscal policy is to ensure fiscal and economic stability. Fiscal and monetary policy coordination is of great importance in the disinflation process. The tools of fiscal policy are tax, spending and borrowing. Correct fiscal policy decisions only can be made based on accurate budget forecasts. Budget forecasts play a critical role in formulating tax policies, determining public expenditure paths, revealing public financing needs and determining growth strategies. Underestimating budget deficits might create incentives to increase government spending. On the contrary, overestimating budget deficits may require increasing tax rates, especially indirect tax rates, which puts extra pressure on inflation.

As far as we know, our study is the first attempt to predict budgetary items in terms of economic classification using deep learning in Türkiye. The contribution of this study is to forecast government spending, tax and their subcomponents using ANN and DNN models, providing important inputs for fiscal management, fiscal planning and fiscal monitoring. Annual data were used and divided into two sub-periods: a training set (2002-2019) and a test set (2020-2022). Each fiscal item is estimated using relevant explanatory variables selected based on economic theory.

We achieved a good forecasting performance for main budget items using ANN and DNN methodologies. We found that most of the Mean Absolute Error (MAE) values fell within the acceptable range, an indicator of good prediction performance. Second, we see that the MAE values of public expenditures are lower than taxes. Third, estimating total tax revenues (aggregate data) performs better compared to subcomponents of taxes (disaggregated data). The opposite is true for public expenditures.

I. Introduction

The application of deep learning in the form of various network models to economics and forecasting in particular is a growing field. Neural network models such as Artificial Neural Network (ANN) and Deep Neural Network (DNN) models have been found to be useful to predict the future values of a variable. These models have been used for the forecasting of several macroeconomic indicators such as GDP growth, inflation, exchange rates and stock returns. It is also common for this methodology to be used in fiscal policy, mainly by estimating government expenditures. Particularly, measuring government size and estimating government expenditures using functional classification of spending are the main focuses of new methodology in the fiscal policy side. Since the composition of public expenditures is important for economic growth, the role of public spending in shaping economic growth is the focus of these studies.

Samimi & Darabi (2011) applied ANN for forecasting government size in Iran. They used annual data covering the 1971-2007 period and selected tax income, oil revenue, GDP per capita, openness and population as explanatory variables for predicting government expenditures. Another example is Magdalena, Logica, & Zamfiroiu (2015), who estimate public expenditures using feed-forward neural networks for five Central and Eastern European countries (Hungary, Poland, Czechia, Bulgaria and Romania). They used annual data and analyzed determinants of the functional distribution of the government spending such as social protection, health, education, general public services and economic affairs. They aim at forecasting main public expenses by functions to create a base for priorities of these countries in terms of economic growth and social welfare. Ghiassi & Simo-Kengne (2021) forecast public expenditures using a machine learning tool (DAN2- a dynamic architecture for ANN) for South Africa. They used quarterly data from 1960 to 2016 and incorporated public expenditures as a percentage of GDP and an old age dependency ratio defined as the ratio of the older population to the working age population. They compared the forecast performance of a neural network model and a traditional time series model (ARIMAX), and they found that the former outperformed the latter. They also highlighted the fact that population aging is an essential predictor of public expenditures. More specifically, the authors report that as population ages, related expenditures increase, which results in higher budget allowances for health and social protection expenses in total budget expenses. Yang et al. (2023) offer another study which shows the superiority of the neural network model (GRU- a gated recurrent unit) compared to other techniques. This study forecast government spending using annual data from 1990 to 2020 for fifteen largest economies in the world.

Deep learning excels at capturing intricate, non-linear relationships without requiring pre-specified functional forms. Unlike classical econometric models such as linear regression or ARIMA, which are linear models, neural networks automatically learn these relationships through hierarchical feature extraction. Specifically, deep learning approaches have been found to reduce forecasting errors and enhance predictive capabilities, achieving higher accuracy rates than traditional methods. While deep learning excels in forecasting, classical econometrics remains superior in causal inference and interpretability. The choice depends on the study: deep learning for forecasting accuracy and econometrics for causal inference. Moreover, deep learning techniques have demonstrated superior performance in various applications compared to classical econometric methods. For example, Bucci (2020) showed that deep learning models can outperform traditional econometric methods in forecasting tasks in financial time series. In addition, Sun (2022) demonstrated that deep learning models can effectively capture complex and non-linear dependencies and thus are more effective in modelling economic and financial data for forecasting purposes, unlike classical econometrics.

The first attempt in Türkiye to estimate the budget allowances of two different institutions, the State Planning Organization and the Court of Accounts, using deep learning is Başaran et al. (2012). They used the ANN method and found that this application provided very accurate public expenditure forecasts for both institutions. We extended this study in terms of estimation technique and data to forecast budget expenditures, tax revenues and their sub-components. We used complex models (ANN and DNN) and a comprehensive (rich) data set to improve the forecast performance of budgetary items.

Modeling and predicting the development of the budget system is difficult and complex because there are many factors that influence the process. This study aims at forecasting the future behavior of output variables, in our case public expenditures, tax revenues and their sub-components, using ANN and DNN techniques for Türkiye. Annual data were used and divided into two sub-periods: a training set (2002-2019) and a test set (2020-2022). Each fiscal item is estimated using relevant explanatory variables selected based on economic theory. Other studies on fiscal policy are also pursued to identify relevant explanatory variables for predicting budget variables.

The aim of fiscal policy is to ensure fiscal and economic stability. Fiscal and monetary policy coordination is of great importance in the disinflation process. The tools of fiscal policy are tax, spending and borrowing. Correct fiscal policy decisions can only be made based on accurate budget forecasts. Budget forecasts play a critical role in formulating tax policies, determining public expenditure paths, revealing public financing needs and determining growth strategies. Underestimating budget deficits might create incentives to increase government spending. On the contrary, overestimating budget deficits may require increasing tax rates, especially indirect tax rates, which puts extra pressure on inflation. The goal of this study is to predict budget items using deep learning, namely ANN and DNN, and contribute to the estimation process. We used the rate of increase in annual data. We expect that forecasting budgetary items using machine learning will improve forecasting performance and play a complimentary role.

The motivation behind this study can be summarized as follows: Firstly, as stated above, sound fiscal policy decisions depend on accurate estimation of budget items. Secondly, the fiscal authority in Türkiye wishes that the budget deficit to GDP ratio remain below 3 percent unless there is an extraordinary situation. In this sense we observe five exceptions, when the country recorded the largest budget deficits in the last twenty years. These years are 2009 and 2010 (due to the global financial crises), 2020 (COVID), and 2023 and 2024 (earthquake expenditures). The budget deficit to GDP ratio, which was 1 percent in 2022, increased to 5.2 percent and 4.9 percent in 2023 and 2024, respectively, with the effect of earthquake expenditures. The existence of high budget deficits in two subsequent years requires authorities to pursue a tight fiscal policy in the following years. And doing so requires accurate estimation of budget revenues and expenditures. Thirdly, increasing budget rigidity reduces the maneuverability of fiscal policy. Regaining flexibility in fiscal policy requires increasing tax audits and taking savings and efficiency into account in public expenditures using artificial intelligence. Lastly, while fiscal policy is affected by macro variables such as inflation and growth, it also affects macro variables. The bi-directional relationship between fiscal variables and macro variables requires accurate estimation of budget items. Since the effects of subcomponents of tax and government spending on national income and inflation vary, accurate estimation of budgetary items gains importance.

As far as we know, our study is the first attempt to predict budgetary items in terms of economic classification using deep learning in Türkiye. The contribution of this study is to forecast government spending, tax and their subcomponents using ANN and DNN models, which are important for fiscal management, fiscal planning and fiscal monitoring.

The outline of this paper is as follows. Section II gives information about the forecast performance of public expenditures and tax revenues included in the Medium-Term Program (MTP). In section III, we explain the features of the methodology (ANN and DNN) that we used in our work. In section IV, we describe data. We present and evaluate the estimation results of the study in section V. Section VI is devoted to robustness analysis. Finally, section VII summarizes the findings of the study and concludes.

II. Forecast Performance of MTP

The government announces the MTP, which presents the next three years' forecasts regarding macro variables (growth, inflation, current account, unemployment, budget deficit, etc.) in September/October each year. The MTP includes the realization forecast of the current year's budget deficit and the forecast for the budget deficits in the following three years. It also contains comprehensive information about fiscal variables (budget deficit, government spending, budget revenues, debt stock, public saving, tax burden, etc.) in Türkiye. It covers forecasts for the central government, general government and public sector borrowing requirements over a three-year perspective.

The forecast performance of budget variables in the MTP can be seen in Figure 1. Growth rates of budget expenditures and tax revenues are displayed in Panel A and Panel B, respectively. We introduce three different definitions to measure forecast performance of fiscal indicators. The definitions are as follows:

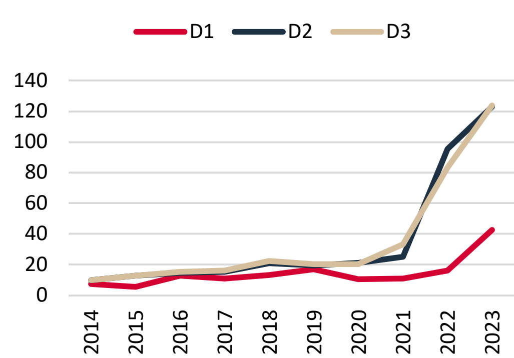
D1: The growth rate of a fiscal variable based on percentage change between estimated value of a fiscal variable in year (t) and its realization forecast in year (t-1)

D2: The growth rate of a fiscal variable based on percentage change between realization forecast of a fiscal variable in year (t) and its actual value in year (t-1)

D3: The growth rate of a fiscal variable based on percentage change between actual value of a fiscal variable in year (t) and its actual value in year (t-1)

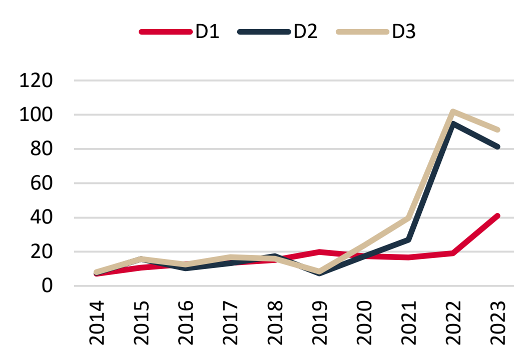
Figure 1: Forecast Performance of MTP

Panel A: Budget Expenditures (% Change)



Source: MTP, Authors' calculations.

Panel B: Tax Revenues (% Change)



Source: MTP, Authors' calculations.

The main conclusion from this analysis is that the forecasting performance of budget variables until 2019 is good, as can be seen from the small differences between the three definitions of the growth rates of fiscal variables. However, we observe a deterioration in budget forecast performance after this year. As seen in Figure 1, the gap between D1 and D2/D3 widens after 2019 for government spending and after 2018 for tax revenues. Even the difference between D2 (comparison of the realization forecast value in the current year and the actual value in the previous year) and D3 (comparison of the actual value in the current year and the actual value in the previous year) increased after 2020 for the former and after 2019 for the latter.[1] The difference between D2 and D3 is expected to be small due to the timing of the realization forecast. The current year's realization forecast, which allows the fiscal authority to observe the budget results of the first eight months, is published in September. Additionally, the government has more information about the general outlook of the economy (growth, inflation, etc.) in September. Since all these factors can positively affect forecast performance, D2 and D3 are expected to be close to each other unless there is an extraordinary shock to the economy.

III. Methodology

Although neural network models have recently attracted a lot of attention and have been used in many fields, their history dates to long before today. The discovery of neurons initiated pioneering studies in the 1950s, which led to the invention of the perceptron, which can be used to model neural networks, and the beginning of deep learning models (McCulloch & Pitts, 1943; Hebb, 1949; Rosenblatt, 1958). Recently, thanks to the improvements in algorithms, data and computational power, neural network models have become available in many areas of application, such as image recognition, text recognition, decision making, anomaly detection and time series.

In recent years, deep learning has also had a significant impact on the field of economics and finance, demonstrating its versatility in various applications. One notable application is in the finance and banking sector, where deep learning models are used to enhance predictive accuracy and model evaluation, providing valuable insights for financial services (Huang, Chai, & Cho, 2020). The banking sector benefits from deep learning applications by implementing various improvements and summarizing relevant techniques to aid future research and practical applications (Hassani et al., 2020). In addition, deep learning methods are employed in financial research areas such as the stock market, marketing, corporate banking, and cryptocurrency. In the realm of financial prediction, deep learning models are utilized to forecast stock and foreign exchange prices, demonstrating significant

1 The year 2023 is an exception for the forecast performance of government spending as seen in the similar growth rates between D2 and D3.

accuracy and return rates (Hu, Zhao, & Khushi, 2021). Furthermore, deep learning has enhanced crisis prediction in fields such as sovereign debt and currency (Fioramanti, 2008; Alaminos et al., 2019). These advances highlight the potential of deep learning to provide more accurate and efficient solutions to complex economic problems. The main source of its success is its ability to analyze the structure of complex relationships in data that have a non-linear structure but do not have a normal distribution (Moreno, 2011).

III.1 ANN Models

An ANN model consists of an input layer, an output layer, a single hidden layer and neurons in this layer. ANNs are single hidden layer computational models inspired by the human brain and have a simpler structure than the DNN models that will be discussed in the next section. The neural network model transforms the input data through a series of linear combinations and non-linear activations to produce a final output. Each neuron receives input from multiple other neurons weighted by parameters known as weights. These weights are adjusted during the training process to minimize the error between the predicted output and the actual target values. The transformation within each neuron is typically performed using an activation function, which adds non-linearity to the model and enables it to capture complex patterns in the data. Figure 2 is an example of an ANN. It contains M input neurons, N neurons in its hidden layer and an output neuron.

Figure 2: The ANN Mechanism

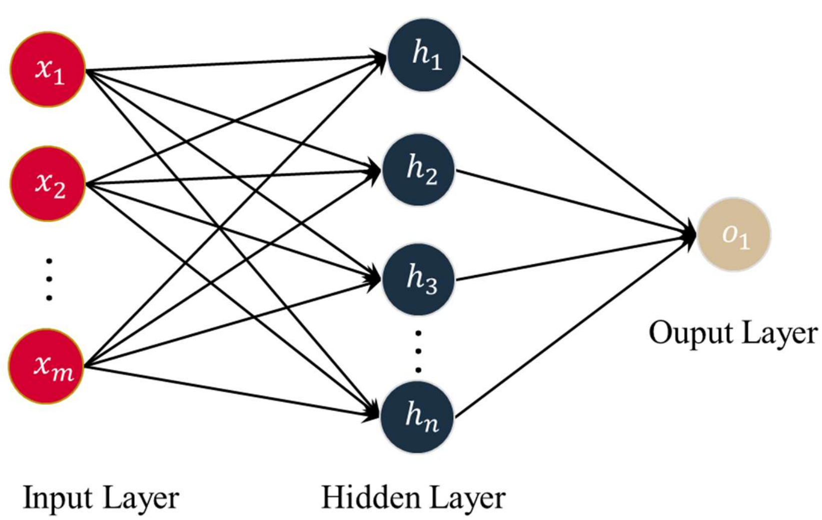


Figure 2 shows how the neurons are connected. Mathematically, this relationship can be represented by defining the pre-activation and activation functions. For neural networks, the pre-activation (Z) of a neuron is the weighted sum of its inputs plus a bias term, represented as

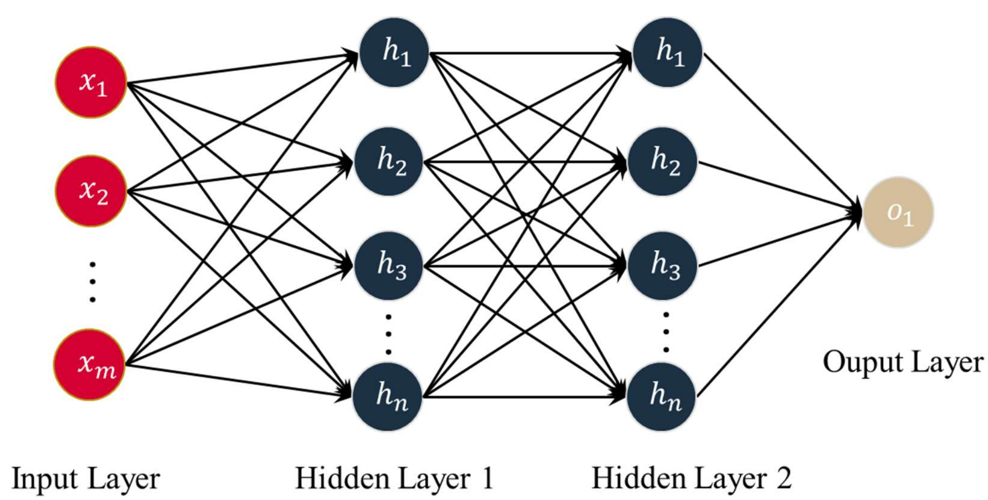
$$z = \sum_{i=1}^{n} w_i x_i + b$$

This pre-activation value is then passed through an activation function to compute the output activation $a(z)$. The activation function introduces non-linearity to the network by transforming the pre-activation value into the final output activation, which is used as the input for the subsequent layer. This process enables the network to learn complex patterns and relationships in the data.

III.2. DNN Models

A DNN model consists of layers and connections and each layer consists of neurons. The number of neurons, the number of layers and the connections between them constitute the architecture. DNNs are computational models inspired by the human brain, consisting of multiple layers of interconnected nodes, or neurons. These networks transform input data through a series of linear combinations and non-linear activations to produce a final output. Each neuron receives inputs from multiple other neurons, weighted by parameters known as weights. These weights are adjusted during the training process to minimize the error between the predicted output and the actual target values. The transformation within each neuron is typically performed using an activation function, which introduces non-linearity into the model and enables it to capture complex patterns in the data. Figure 3 shows us an example of a DNN with two hidden layers. It contains M input neurons, N neurons in its hidden layers and one output neuron.

Figure 3: The DNN Mechanism



Training a DNN involves optimizing the weights using algorithms like gradient descent and its variants. During training, the network's output is compared to the ground truth, and the error is propagated back through the network using a technique called backpropagation. Backpropagation computes the gradient of the loss function with respect to each weight by applying the chain rule of calculus, allowing the weights to be updated in a direction that minimizes the error. The iterative process of forward passes, where inputs are transformed through the network to produce outputs, and backward passes, where gradients are calculated and weights are updated, continues until the model converges to an optimal set of weights.[2]

III.3. Train-Test Split

The train-test division is an important process for successful results in machine learning and deep learning. Especially with the out-sample test, it is a key process for measuring the error margin of the model in previously unseen inputs and for the success of the model. In the test-train split process, the data dimension was considered and the last three-year period for the test was determined.

Why do we use full-batch rather than mini-batch? Using full-batch gradient descent in DNN ensures stable and accurate convergence by averaging gradients over the entire dataset, which reduces variance and enhances reproducibility. This method simplifies theoretical analysis, making it easier to benchmark and validate findings. It is particularly effective for small datasets that fit into memory, as it eliminates the need for data shuffling and batching, leading to efficient training and precise parameter updates, which help in mitigating overfitting and fine-tuning the model.[3]

III.4. Activation Function and Algorithm

The Rectified Linear Unit (ReLU) activation function is widely used in DNN due to its simplicity, computational efficiency, and effectiveness in mitigating common training issues. ReLU introduces non-linearity, enabling the network to learn complex patterns and representations, while its simple operation (replacing negative values with zero) speeds up computation and training. Additionally, ReLU promotes sparse activation, improving computational efficiency and reducing overfitting. These benefits make ReLU a powerful and popular choice in deep learning applications (Agarap, 2018; Bai, 2022; Simpson, 2015; Cui et al., 2017).

[2] One may wonder whether neural networks explicitly follow a general-to-specific reduction scheme where irrelevant variables are systematically eliminated, as in econometrics. During the gradient-based learning, variables with little or no contribution to the prediction are assigned smaller weights, which may effectively diminish their influence on the output. Moreover, the optimal weights in a neural network are not directly analogous to econometric coefficients in terms of interpretability. Econometric coefficients are designed to capture the marginal effect of an explanatory variable on the dependent variable, holding all other variables constant. In contrast, NN weights are part of a complex network structure, where each weight contributes non-linearly to the output.

[3] Neural networks can inadvertently learn from irrelevant or less relevant variables, especially in the presence of noise, overfitting or high dimensionality. However, various mechanisms such as variable selection help mitigate this risk and reduce the likelihood of irrelevant features being included, model architecture selection naturally focuses on the most relevant regions of the input space, properly designed loss functions can emphasize prediction accuracy while penalizing over-fitting to noise.

$$R(z) = \begin{cases} z & z > 0 \\ 0 & z \leq 0 \end{cases}$$

The Adam optimization algorithm is preferred for training neural networks because it combines the advantages of two other popular optimization methods: AdaGrad and RMSProp. Adam adapts the learning rate for each parameter, improving performance on problems with noisy or sparse gradients, and requires less memory than other adaptive methods. It also converges quickly by maintaining a balance between the benefits of per-parameter learning rates and the efficiency of momentum, making it particularly effective for large datasets and high-dimensional parameter spaces (Kingma & Ba, 2014; Wang et al. 2018).

Mean Absolute Error (MAE) and Mean Squared Error (MSE) are common metrics used to evaluate the accuracy of DNN models in time series forecasting. MAE measures the average magnitude of the errors in a set of predictions, without considering their direction, providing an intuitive sense of the average prediction error. It is particularly useful when dealing with non-Gaussian distributions or when outliers need to be less influential. MSE, on the other hand, calculates the average of the squared differences between predicted and actual values, emphasizing larger errors due to its squaring effect. This makes MSE more sensitive to outliers, which can be beneficial if large errors are particularly undesirable. Both metrics have their applications: MAE is often preferred for its simplicity and interpretability, while MSE is useful for highlighting larger errors and providing more mathematical convenience in optimization problems (Chai & Draxler, 2014; Hodson, 2022; Willmott & Matsuura, 2005).

$$MAE = \frac{1}{n}\sum_{i=1}^{n} |Y^{Pred} - Y^{Actual}|$$

$$MSE = \frac{1}{n}\sum_{i=1}^{n} (Y^{Pred} - Y^{Actual})^2$$

III.5. Hyperparameter Tuning and Post-Training Simulations

Since it is crucial to find the right and consistent architecture for sharp and accurate predictions, we defined a search space. We explored all combinations of parameters in the search space and found the architecture that gave the best results. While finding the best architecture, multiple simulations were performed for each combination and the error scores were averaged and compared[4]. Below are the components of the explored search space.

[4] Within the scope of the study, Python 3.11 was used in the data process and deep learning calculations were calculated with graphics processing units (GPU). Thus, computations requiring high computational power were parallelized and efficiency was increased.

$$Neuron\ Space = \{8, 16, 32, 64, 128, 256, 512\}$$

$$Layer\ Space = \{1, 2, 3, 4, 5\}$$

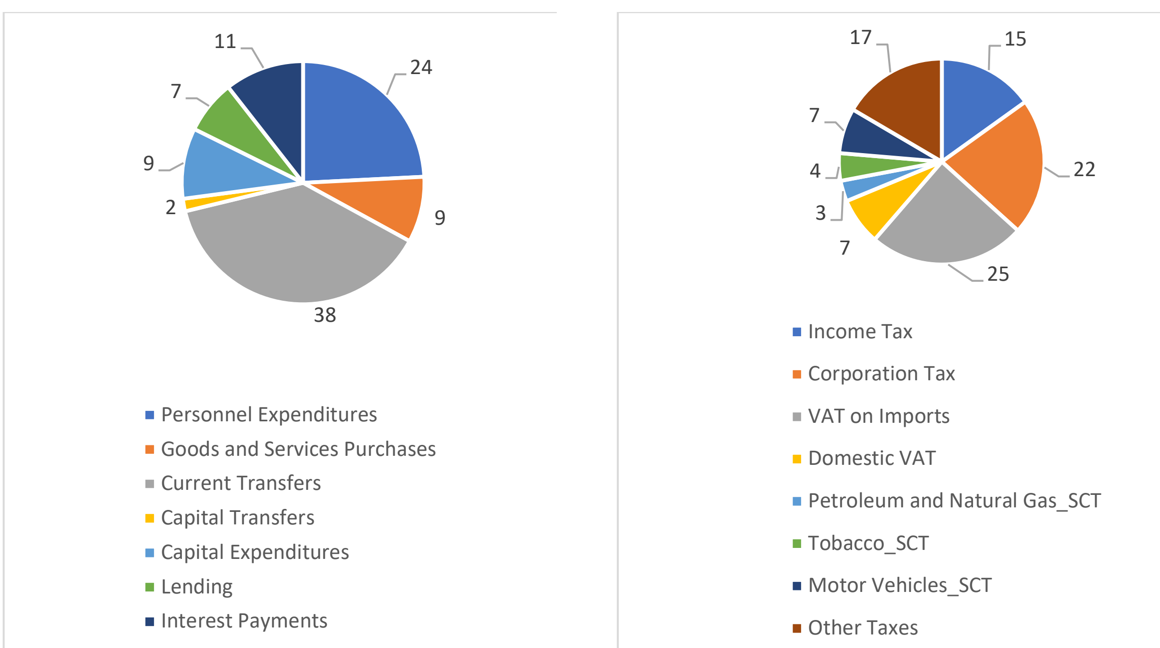
After the best architecture was found, the model trained using training data was estimated with test data and the accuracy and consistency of the results were examined. The consistent predictions obtained as a result of 100 simulated trials were compared with the actual values.

IV. Data

Before giving information about the data used in this analysis, it would be useful to clarify the structure of the budget items. One can see the components of the budget items in Figure 4. The share of each component of government spending in total spending in 2022 is shown in Panel A. Panel B illustrates the share of each component of tax in total tax revenues in 2022. According to Panel A, current transfers constitute 38% of total government spending, followed by personnel expenses with 24% and interest payments with 11%. Panel B shows that value added tax (VAT) on imports constitutes 25% of total tax revenues, followed by corporation tax with 22% and income tax with 15%.

Figure 4: Composition of Budget Items in 2022

Panel A: Share of Budget Expenditures (%) **Panel B: Share of Tax Revenues** (%)



Source: Ministry of Treasury and Finance of Türkiye

We used percentage change in annual data in the analysis. The data covers the period 2002-2022[5]. Central government budget expenditures and tax revenues are collected from the Ministry of Treasury and Finance. We estimated both aggregated (budget expenditures and tax revenues) and disaggregated (their main components) data by using relevant explanatory variables for each item. On the expenditure side, personnel expenditures, good and service purchases, current transfers, capital expenditures, capital transfers, lending and interest payments (both domestic and external) were taken into account. Regarding public revenues, the main tax revenue items selected are income tax, corporation tax, special consumption tax (SCT), domestic VAT and VAT on imports for the estimation process[6]. All data used in the analysis are summarized in Table 1. The data other than budget items were received from other institutions including the Central Bank of the Republic of Türkiye, the Turkish Statistical Institute, the Social Security Institution, the Energy Market Regulatory Authority, the Banking Regulation and Supervision Agency, the Presidency of Strategy and Budget, the Ministry of Agriculture and Forestry, and the Automotive Manufacturers Association.

Table 1: Data

Predicted Variables	**Explanatory Variables**
Income Tax	Gross wage salary index, deposit interest rate, number of employees
Corporation Tax	Bank profits, exchange rate, growth, Inflation, number of taxpayers, tax rate
VAT on Imports	Imports excluding gold, exchange rate, tax rate
Domestic VAT	Growth, inflation, exchange rate, revenues from SCT
Petroleum and Natural Gas_SCT	Fuel sales, lump-sum tax (gasoline, diesel and LPG)
Tobacco_SCT	Cigarette sales, cigarette price, lump-sum tax, proportional tax
Motor Vehicles_SCT	Automobile sales, light commercial vehicle sales, automobile price
Personnel Expenditures*	Public employment, civil servant salary
Goods and Services Purchases	Growth, Inflation, other primary public expenditures, budget revenues
Current Transfers	Growth, inflation, exchange rate, number of retirees, retirement pension Other primary public expenditures, tax revenues, natural gas price, oil price
Capital Transfers	Growth, inflation, other primary public expenditures, budget revenues
Capital Expenditures	Growth, inflation, other primary public expenditures, budget revenues
Lending	Growth, Inflation, exchange rate, other primary public expenditures budget revenues, natural gas price
Dom. Debt Interest Payments	Domestic borrowing interest rate, exchange rates (euro and US dollar), CPI gold price, domestic debt stock
Foreign Debt Interest Payments	Eurobond interest rate, exchange rates (euro and US dollar), foreign debt stock

* Personnel expenditures consist of compensation of employees and social security contributions

5 We chose the year 2022 as a last year because of earthquake expenditures occured in the period of 2023-2024. Largely discretionary nature of earthquake expenditures deteriorates the quality of estimation. On the other hand, the mandatory tax adjustments for earthquake expenditures are of an optional nature, making it difficult to make accurate estimates without removing these temporary elements from the budget items.

6 Tax revenues constituted 84% of total budget revenues in 2022. Non-tax revenues covering property income, interest, shares and fines, capital revenues etc. are outside the scope of this study.

Selected sub-items of tax revenues constitute the most significant part (approximately 60%) of total tax revenues. We chose the two main components of direct tax revenues, namely income tax and corporation tax. While gross wage salary index, number of employees and deposit interest rate were selected for income tax, bank profits, number of taxpayers, tax rate and macro variables (growth, inflation, exchange rate) were added for corporation tax. As for indirect tax revenues, we chose VAT and SCT. We divided VAT into two components: domestic VAT and VAT on import, and SCT into three main components: petroleum and natural gas, tobacco and motor vehicles. As for domestic VAT, growth, inflation and the exchange rate were included as well as revenues from SCT as explanatory variables[7]. Since our tax system allows taxes to be collected on other taxes, SCT revenues are included in the tax base of domestic VAT. In calculating VAT on imports, imports excluding gold, exchange rate and tax rate were used. SCT on petroleum and natural gas was estimated using fuel sales and lump-sum tax on fuels. Since most of the tax is generated from automotive fuels, we included gasoline, diesel and LPG sales and their lump-sum taxes in the analysis. Similarly, SCT on tobacco was estimated by taking into consideration cigarette sales, cigarette prices and taxes on the cigarettes (both proportional and lump-sum taxes). SCT on automobiles was also calculated using automobile sales, light commercial vehicle sales and automobile price.

Although it is quite common to use functional classification for estimating government spending in the literature, we used economic classification to estimate expenditure items for Türkiye. The number of public employees and civil servant salaries were used to estimate personnel expenditures which constitute the most important part of government consumption. Purchases of goods and services, which constitute the remaining part of government consumption, were estimated by taking into account growth, inflation, other non-interest public expenditure[8] and budget revenues. The reason for including the last two variables in the prediction can be explained as follows: the more taxes the government collects, the more it spends. Moreover, the size of other non-interest expenditure limits the government from spending more on the relevant expenditure item (in this case, goods and services purchases). We also used the same variables for government investment (capital expenditures and capital transfers). We estimated current transfers, the largest component of public expenditures, using macro variables (growth, inflation, exchange rate) and commodity prices (natural gas and oil prices), pension and the number of retirees and budgetary items such as tax revenues and other primary public expenditures.

[7] Alternatively, we also used the growth rate of final consumption expenditure of resident households instead of GDP growth. This choice did not improve prediction performance much. We also include the change in credit card stock to account for domestic VAT. Similarly, improvement remains limited.

8 Other non-interest (primary) expenditures refer to total non-interest (primary) expenditures excluding estimated related budget expenditures.

Interest payments were divided into two parts: domestic and foreign. We chose explanatory variables for each item by taking into consideration the composition of debt stock and type of borrowing. Therefore, domestic interest payments were estimated using Treasury domestic borrowing interest rates, exchange rates (euro and US dollar), CPI, gold price and domestic debt stock. On the other hand, Eurobond interest rate, exchange rates (euro and US dollar) and foreign debt stock were used to forecast foreign debt interest payments.

Alternatively, we also estimated aggregate variables (total government spending and tax revenues) using the whole dataset for each aggregate item. The dataset was divided into two parts: a training set (period 2002-2019) and a test set (2020-2022). As seen in Section VI, alternatively, we shortened the training period (2002-2016) and applied the testing period as 2017-2019. By doing this, we enable the model to realize its power by taking into account the pre-COVID period.

V. Results

This section is allocated to interpret the results of the study. Evaluation of prediction results is based on mean absolute error (MAE) values[9]. Since we use the percentage change of variables, it will be useful to evaluate the performance of the model using MAE values. Table 2 shows MAE values for each item. The lower the MAE, the better the model predicts.

Table 2: Model Results

Tax Revenues	**MAE**	**Budget Expenditures**	**MAE**
Income Tax	6.1	Personnel Expenditures	1.0
Corporation Tax	6.6	Goods and Services Purchases	2.6
VAT on Imports	7.7	Current Transfers	2.9
Domestic VAT	25.0	Capital Expenditures	1.5
Petroleum and Natural Gas_SCT	5.4	Capital Transfers	20.4
Tobacco_SCT	5.1	Lending	28.0
Motor Vehicles_SCT	50.9	Domestic Debt Interest Payments	2.6
		Foreign Debt Interest Payments	4.2
Tax Revenues	3.3	Budget Expenditures	3.0

We choose four budget variables to investigate the results in depth. These variables are purchases of goods and services, current transfers, total government spending and total tax revenues. Forecast results are provided in both tabular and graphical forms. Figure 5,6,7 and 8 display actual values and estimated values for each budget item in terms of % change and level. Tables 3 shows the three-year MAE values for each variable along with the corresponding actual and estimated values. Focusing on budget expenditures, it turns out that the estimated values are close to the actual values, allowing

9 The actual values of the explanatory variables in the current year were used in the estimation of budget items for the forecast period (2020-2022).

average MAE values to be around 3[10]. In the first year it is 4, in the second year it is 1.84, and in the third year it is 3.23. It is acceptable to obtain MAE values under 10, and we get 3 in our study. For example, in the second year while the actual value is 33.21 percent, the estimated value is 35.06 percent. We also track the better performance in Figure 5. Panel A shows that while the model over-estimates the increase rate of government spending in the first and second year, it under-estimates the increase rate in the third year. This balancing process appears to help accurately estimate the level of budget expenditures (Figure 5, Panel B).

Table 3: Comparison of Actual and Estimated Values

	Budget Expenditures			**Tax Revenues**		
	Actual Values	Estimated Values	MAE	Actual Values	Estimated Values	MAE
Year 1	20.37	24.37	4.00	23.65	26.39	2.74
Year 2	33.21	35.06	1.84	39.81	33.41	6.40
Year 3	83.52	80.29	3.23	102.01	101.18	0.83
Average			3.02			3.32
	Current Transfers			**Goods and Services Purchases**		
	Actual Values	Estimated Values	MAE	Actual Values	Estimated Values	MAE
Year 1	24.42	24.50	0.08	14.95	17.75	2.81
Year 2	25.85	33.91	8.06	37.62	33.25	4.38
Year 3	79.69	79.20	0.49	93.07	92.36	0.71
Average			2.88			2.63

We observe the same story in tax revenues. MAE values are calculated as 2.74, 6.4 and 0.83 in 2020, 2021 and 2022, respectively (Table 3). Although its forecast performance was not good in the second year, it made up for it in the third year. Therefore, the average MAE value of these three years is 3.32[11]. As shown in Figure 6, the change in tax revenues and the level of tax revenues are estimated well. Consequently, we find a better fit for aggregate budget variables, government spending and tax revenues.

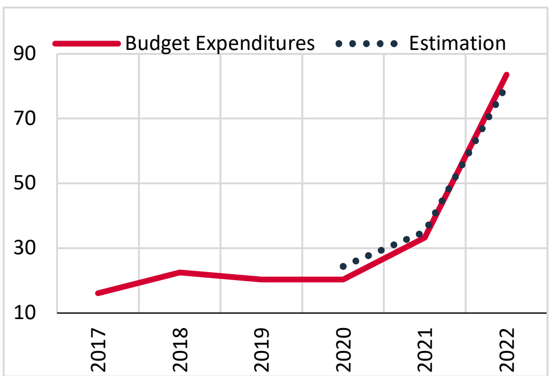
We also investigate the forecast performance of two subcomponents of public expenditure. One of these is the purchase of goods and services, which are a part of public consumption, and the other is current transfers, which constitute the most important part of total public expenditures. Table 3 shows that both variables have MAE values below 3 and exhibit good predictive performance. The average MAE values over three years are 2.63 and 2.88 for purchases of goods and services and current transfers, respectively. The estimation results reveal that current transfers increased by 24.42 percent in 2020, it was estimated as 24.5 percent, and the difference between the two values is only 0.08.

10 The average MAE value of budget expenditures in the MTP was calculated as 6.9 for the period of 2020-2022. (Based on definition D2 and D3, see section II).
11 The average MAE value of tax revenues in the MTP was calculated as 8.8 for the period of 2020-2022. (Based on definition D2 and D3, see section II).

However, the prediction performance of the model in 2021 dropped sharply, and the difference between the actual and predicted values reached 8.06. In other words, the model over-estimated the speed of current transfers in the second year and the third year estimation value captured the actual one, showing a good predictive performance as in the first year. As a result, the level of estimated current transfers remains slightly above the actual line (Figure 7, Panel B). Moreover, we find that both the growth rate and level of goods and services purchases are estimated very well (Figure 8).

Figure 5: Estimation of Budget Expenditures

Panel A: Budget Expenditures (% Change)



Panel B: Budget Expenditures (Level, Billion TL)

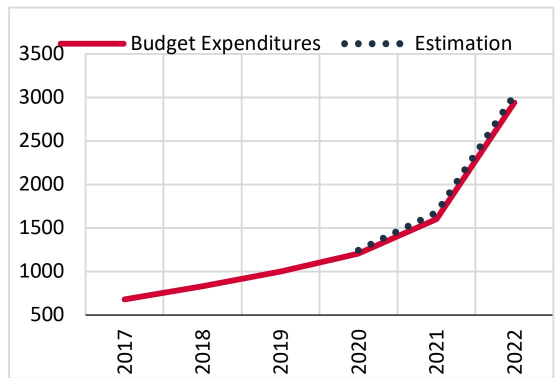
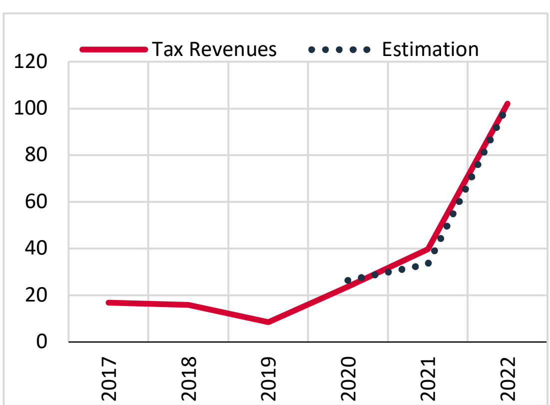


Figure 6: Estimation of Tax Revenues

Panel A: Tax Revenues (% Change)



Panel B: Tax Revenues (Level, Billion TL)

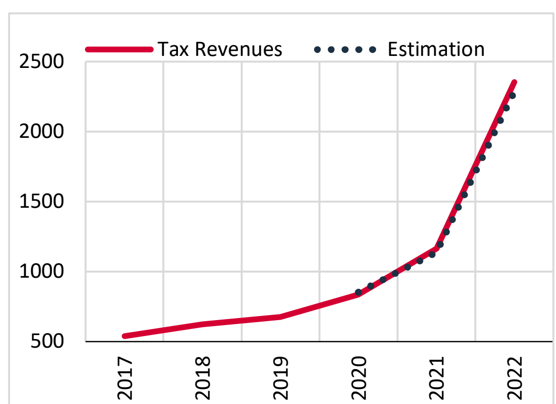


Figure 7: **Estimation of Current Tansfers**

Panel A: Current Transfers (% Change) **Panel B: Current Transfers** (Level, Billion TL)

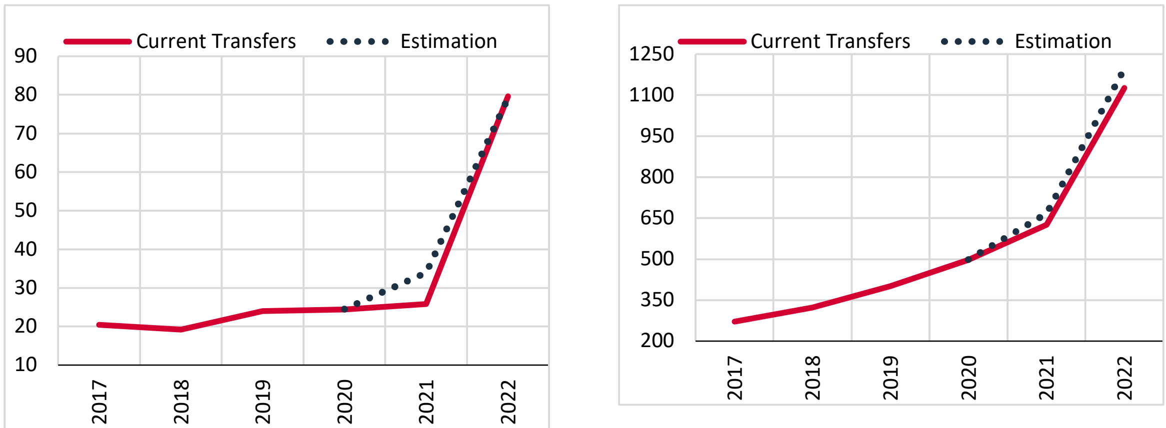
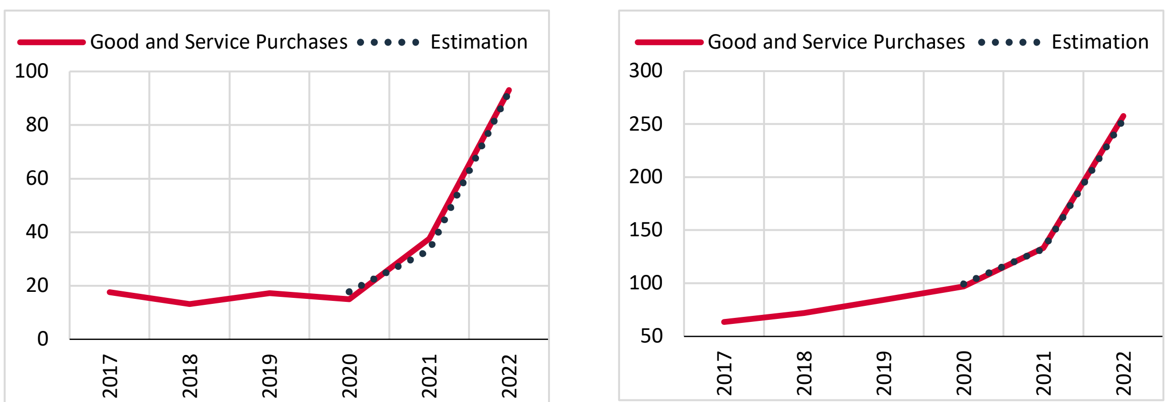


Figure 8: Estimation of Goods and Services Purchases

Panel A: Goods and Services Pucrhases (% Change) **Panel B: Goods and Services Purchases** (Level, Billion TL)



This study also covered the other sub-items. Using disaggregated data might improve the forecast performance of the model. To do this, we model eight types of government spending and seven types of tax revenues. On the expenditure side, we estimated all components of government spending, personnel expenditures, goods and services purchases, current transfers, capital transfers, capital expenditures, lending, domestic debt and foreign debt interest payments. On the revenue side, we selected seven main taxes: income tax, corporate tax, VAT on imports, domestic VAT, special consumption tax on petroleum and natural gas, tax on tobacco, and motor vehicle special consumption tax (SCT). All MAE values of these variables are tabulated in Table 2. We find that the MAE values of public expenditures are lower than they are for taxes. Government spending items that gave the lowest MAE values are personnel expenditures (1.0), and capital expenditures (1.5). These items are followed by goods and services purchases (2.6), domestic debt interest payments (2.6), current transfers (2.9) and foreign debt interest payments (4.2). The worst results relate to capital transfers (20.4) and lending (28.0), which show a variable change on an annual basis. Overall, budget expenditures gave a value of 3 for MAE, which is an indicator of good forecast performance. On the revenue side, special consumption tax on petroleum and natural gas and tobacco gave a good result in terms of MAE values (5.1 for tobacco and 5.4 for petroleum and natural gas). Although these values are around 5 and higher than MAE values of government spending, they are still in an acceptable range. Focusing on direct tax revenues, income tax and corporate tax have MAE values around 6 (Table 2). VAT on imports has relatively high MAE values but it is still acceptable (below 10). On the other hand, interestingly, we obtained a high MAE value for domestic VAT (25.0) and SCT on motor vehicles (50.9). We found 3.3 for the MAE value of total taxes, for which we used all explanatory variables related to taxes. These results reveal the fact that the model estimated total tax revenues (aggregate data) better than its sub-components (disaggregate data). On the other hand, we found that MAE values in the subcomponents of total public expenditures were lower than in total public expenditures. Finally, we found that most budget items had appropriate MAE values that fell within the acceptable range. The only exceptions are capital transfers and lending on the expenditure side, and domestic VAT and SCT on motor vehicles on the revenue side. They have high MAE values which are outside the acceptable range.

VI. Robustness Analysis

This section is allocated to robustness analysis. In this framework we applied a different approach to the estimate model. As mentioned in section III, the standard model uses relevant variables for each budgetary item and gets one output in each case. That is, we estimated each budget variable separately. Alternatively, it is also possible to include all explanatory variables in the model at the same time and get several outputs (in our case 17). This also requires establishing a new architecture. We called the new version of the model a multi-output model and Table 4 illustrates the MAE values of this multi-output model in comparison with the results of the standard model. As shown in Table 4, the standard model performs better than the new version. For instance, while the MAE value for tax revenues is 3.3 in the standard model, it approaches 10 (about three times higher) in the multi-output model. This argument is also valid for other budget variables such as government spending. While the MAE value for government spending is 3 in the standard model, it jumps to 5.8 (about two times higher) in the multi-output model. Therefore, we conclude that estimating each variable separately outperforms the multi-output model which uses all explanatory variables at the same time and obtains several outputs at once.

Table 4: Robustness Analysis-MAE Values of Different Models

	Standard Model	Lagged Model	Multi-Output Model
Budget Expenditures	**3.0**	**6.1**	**5.8**
Personnel Expenditures	1.0	21.7	13.3
Goods and Services Purchases	2.6	24.7	7.2
Current Transfers	2.9	4.3	10.9
Capital Expenditures	1.5	31.4	13.2
Capital Transfers	20.4	19.3	38.3
Lending	28.0	75.6	62.8
Domestic Debt Interest Payments	2.6	26.7	26.4
Foreign Debt Interest Payments	4.2	11.4	23.0
Tax Revenues	**3.3**	**21.8**	**9.7**
Income Tax	6.1	25.0	12.5
Corporation Tax	6.6	61.4	40.7
VAT on Imports	7.7	48.9	10.0
Domestic VAT	25.0	29.8	28.4
Petroleum and Natural Gas_SCT	5.4	55.5	53.3
Tobacco_SCT	5.1	13.9	25.2
Motor Vehicles_SCT	50.9	134.0	95.5

As a second robustness check, we estimated future values of budget items using lagged values of explanatory variables to measure the forecast performance of the model. In the standard model, we used the realization of explanatory variables to receive forecast values of budget variables. In this section, previous year values of macroeconomic variables are used instead of current year realization values. Comparison of the results of the two models (standard and lagged models) shows that using past values of explanatory variables worsens the forecast performance of the model. This situation varies from one budgetary item to another. For example, while the forecast performance of total public expenditure worsens by a limited amount in terms of MAE values (from 3 to 6.1), the gap between the two models expands for total tax revenues (from 3.3 to 21.8) (Table 4). We also observe a similar picture between the two models in budget sub-items.

One may also wonder whether the COVID period affects the forecast performance of the model. To test this, we also estimated a model for the period of 2002-2016 as a training period and 2017-2019 as the test period[12]. In doing so, we exclude the COVID period (2020-2022) and ignore the implementation of expansionary fiscal policy (health expenditures, tax reductions, tax deferrals and transfers to households and firms) in that period. The findings of the study support the idea that changing the estimation and testing period has a negligible effect on our results.

[12] In the standard model, the last three observations corresponding the years between 2020 and 2022 (COVID-period) were used for the test set and the rest of the observations were used for the training set.

VII. Conclusion

We built models (ANNs and DNNs) to forecast the main budgetary items using publicly available data on the Turkish economy. The annual data covers the period of 2002-2022. ANN and DNN models were applied to estimate budget expenditures and tax revenues and their main components. The explanatory variables chosen for each budget item vary according to economic theory. However, when estimating the aggregate budget item, we include all explanatory variables affecting the tax and expenditure subcomponents.

One can summarize the findings of this study as follows:

First, we achieved good forecasting performance for main budget items using ANN and DNN methodologies. We found that most of the MAE values fell within the acceptable range, an indicator of good prediction performance. Second, we see that the MAE values of public expenditures are lower than taxes. Third, estimating total tax revenues (aggregate data) performs better compared to subcomponents of taxes (disaggregated data). The opposite is the case for public expenditures.

One of the caveats of this study is to use macro data for forecasting. Of course, if high frequency data is available, it is possible to extend this study to include micro data, which could improve the model's predictive performance.

For forecasting purposes, we use economic classification of budget data. It is quite common in the literature to use functional classification of budget expenditures. Therefore, it would be useful to estimate government expenditures within the scope of productive and unproductive expenditures to clarify policy implications in terms of economic growth and social welfare. We left this issue for future research.

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